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Unit 2 Bringing Ideas to Life : MODI Automatic Door

PDF Documents

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UNIT 2 BRINGING IDEAS TO LIFE

Project Introduction

TIME OF FLIGHT (TOF) SENSORS IN DAILY LIFE

Time of Flight (ToF) sensors are versatile and increasingly used in various everyday applications due to their ability to accurately measure distances and map environments. Here are some common examples of ToF sensors in daily life:

Smartphones

Many modern smartphones incorporate ToF sensors to improve the autofocus capabilities of cameras, particularly in low-light conditions. They are also used for advanced facial recognition systems that enhance security features by accurately mapping the user's face in three dimensions.

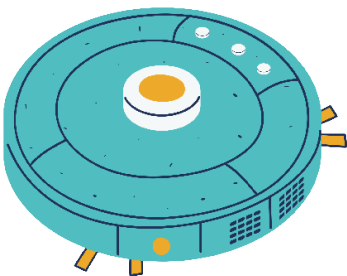


Autonomous Vehicles

ToF sensors are critical in the development of autonomous driving technologies. They help cars detect and avoid obstacles, map environments, and maintain safe distances from other vehicles, thereby enhancing road safety.

Augmented Reality (AR)

In AR applications, ToF sensors are used to enhance the interaction between real-world and virtual elements by accurately measuring the distances and dimensions of the surrounding space. This helps in placing virtual objects in real environments more precisely, improving the user experience in AR games and applications.



Robotic Navigation

Robots equipped with ToF sensors can navigate complex environments more effectively. These sensors help robots avoid collisions, understand the layout of the surroundings, and perform tasks like delivering items within buildings or navigating busy warehouse floors.

Smart Home Devices

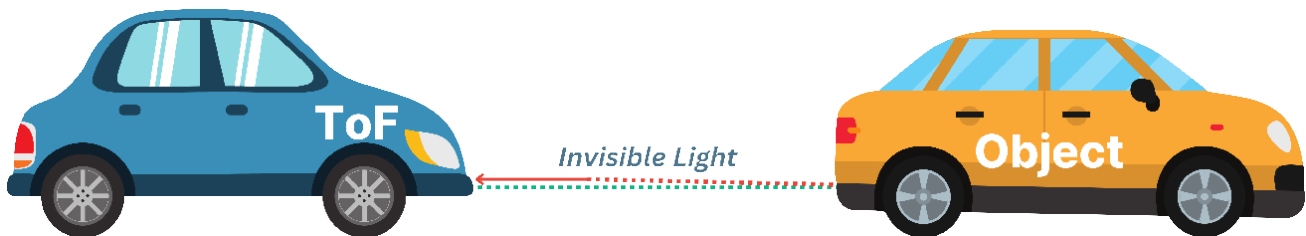
In smart homes, ToF sensors are used in automatic lighting systems, where lights turn on or adjust based on the presence and location of a person. They are also used in advanced home security systems to detect intrusions or unusual movements by measuring distances and identifying obstacles.



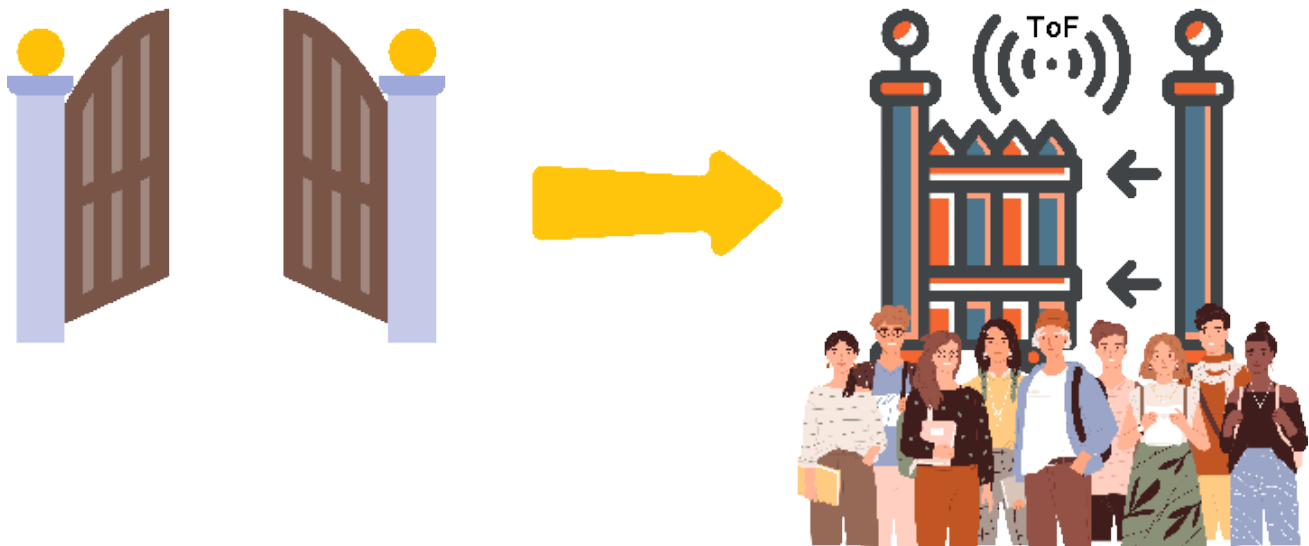
STORY TELLING - DIGITAL ROCK-PAPER-SCISSORS GAME

At Lux High, the students noticed that their modern school still had old, heavy doors. This gave the robotics club a great idea. Why not make a door that opens by itself?

Noah, the leader of the club, used special parts called MODI modules, especially the ToF module, to build an automatic door. He had once read about ToF (Time of Flight) sensors and was fascinated by how they work.



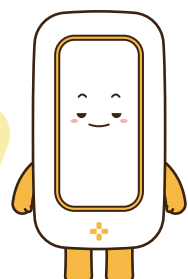
These sensors measure distance by emitting invisible light; when objects block this light, the sensor calculates the time it takes for the light to reflect back to know how far away the object is.



This smart door could sense when someone was coming and open all by itself. Everyone at school loved using the new door, and it showed how even a small project could make a big difference.

TIPS

ToF sensors are used to measure or sense distance. It emits invisible light, and if there is something blocking the light, it calculates the reflection time to get the distance.





Required Modules & Brainstorming

Based on the story, which MODI modules do you need for this creation?



Dial



Button



Environment



Joystick



ToF



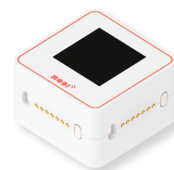
Speaker



IMU



LED



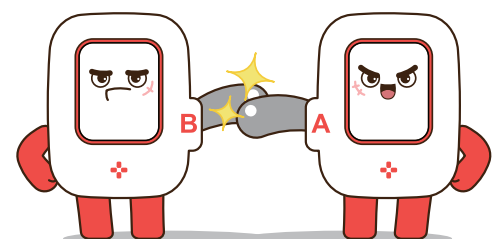
Display



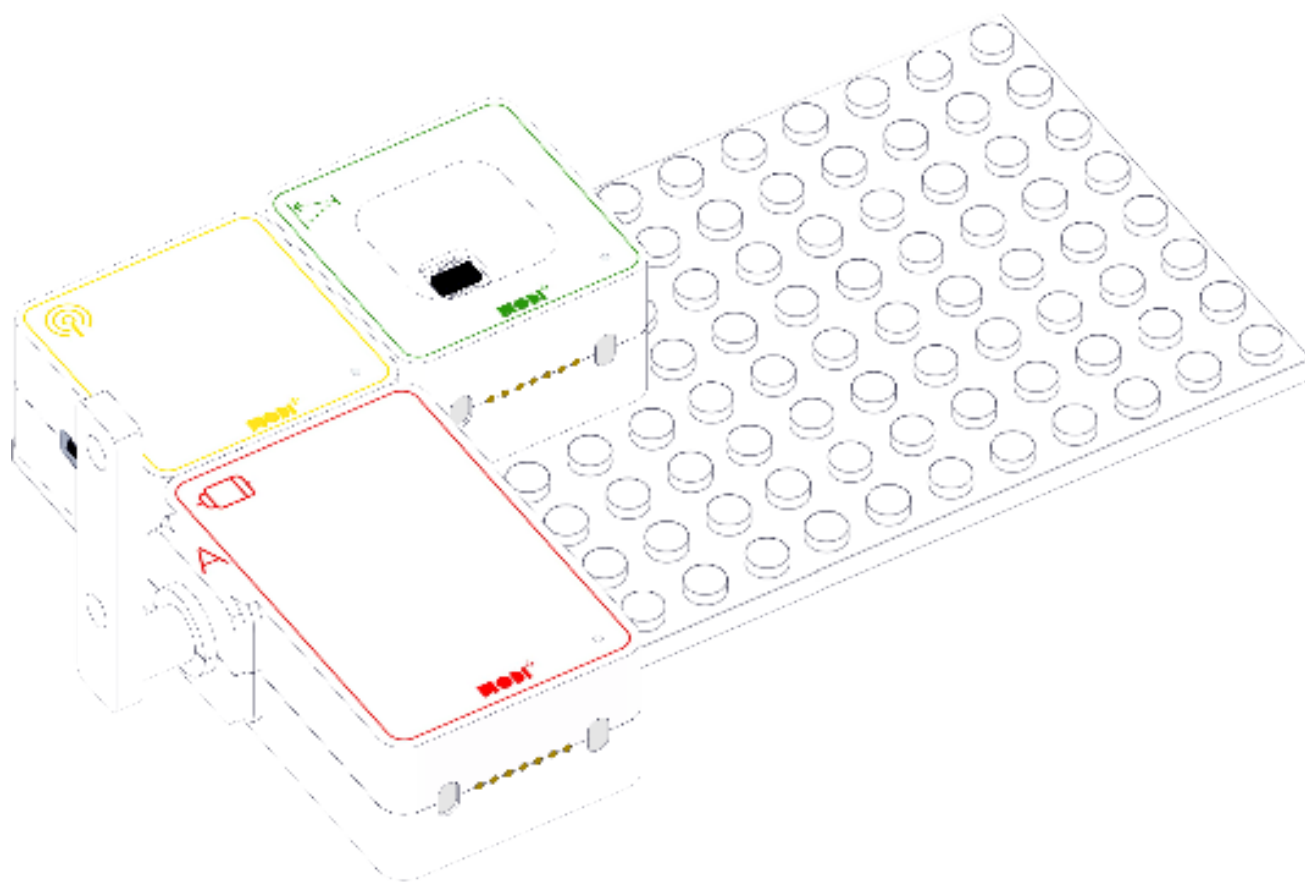
Motor A



Motor B



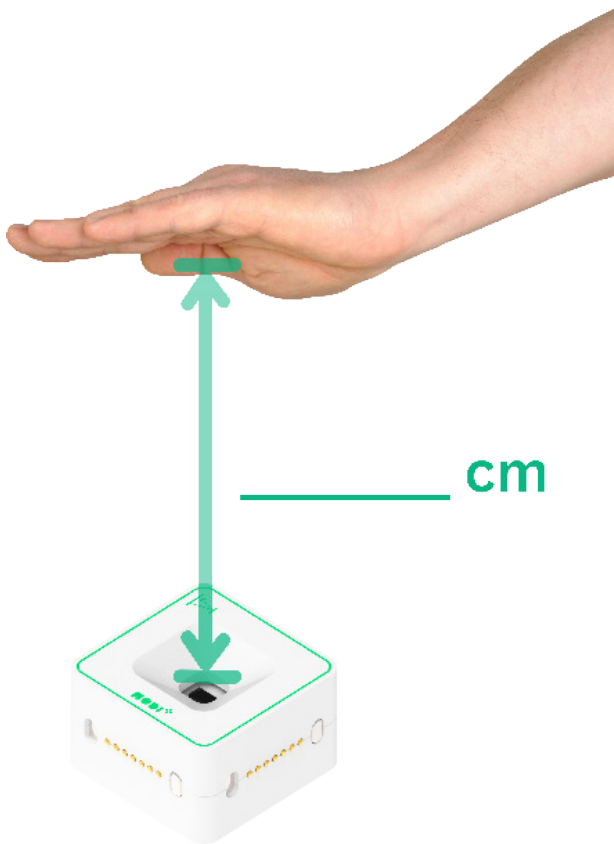
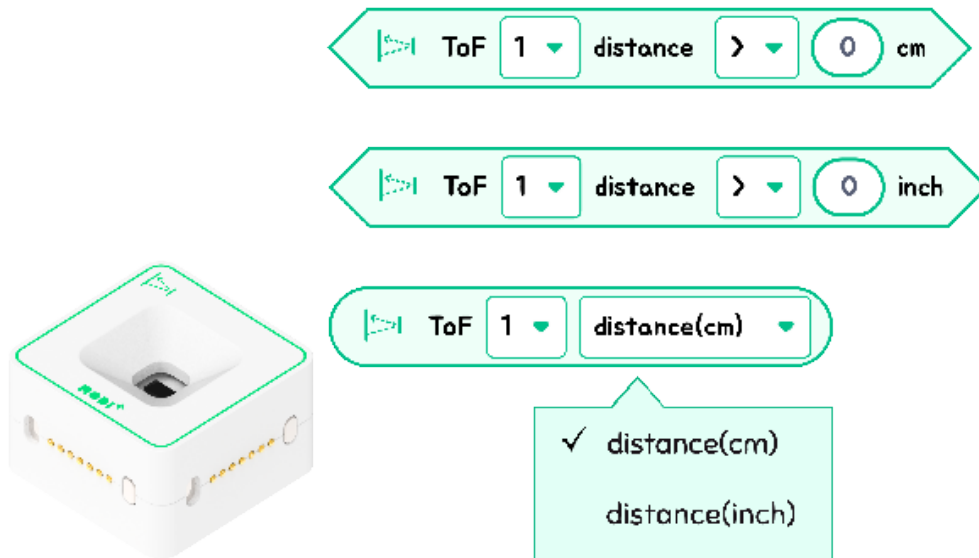
HOW DOES THE CREATION LOOK LIKE?





Module Functionality

Let's look into the details of the ToF module.



Main Features of ToF

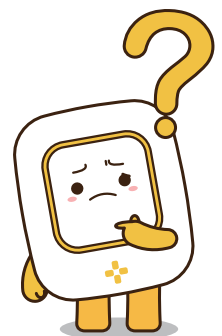
Distance Measurement

It measures the distance to an object by timing how long it takes for a light signal to travel to the object and back.

Real-time Processing

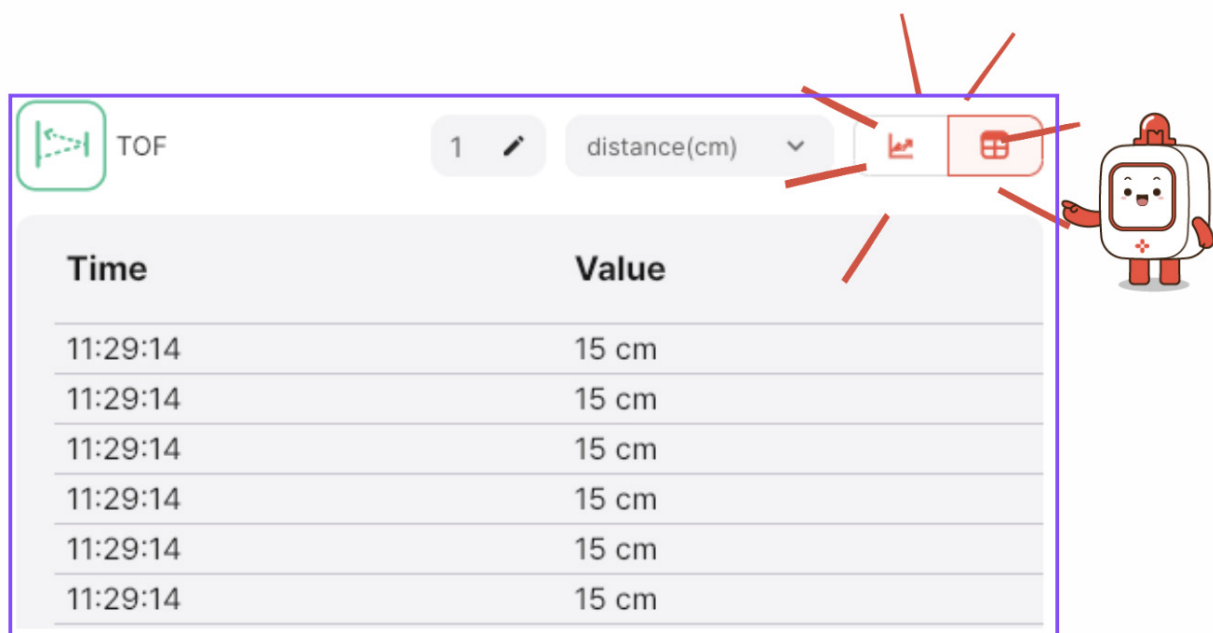
Capable of measuring distances in real-time, which is essential for dynamic environments and applications like robotics and autonomous vehicles.

What is the distance between the ToF module and you hand now?



Monitoring Features:

1. Switch between chart and graph views.



2. The ToF module can monitor distance, providing measurements in centimeters (cm) and inches (inch).

